University of Science and Technology
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CSAI 205: Fundamentals of Circuits and Electronics
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Project 1

Diode-Based Logic Gate Circuits & Multi-Output Power Supply

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1. Objective

The goal of this project is to build the basic logic functions (OR, AND, NOR and NAND) using only diodes, resistors, push buttons and LEDs, without any logic ICs. Each circuit was first simulated in Proteus and then built on a breadboard. The second part of the project is a multi-output power supply.

Through this project we practice reading a circuit diagram, understanding how a diode behaves in forward and reverse bias, and comparing simulation results with real measurements.

2. Components Used

Component	Quantity
Breadboard (test board)	1
Silicon diodes (1N4001)	8
5 mm LEDs (mixed colors)	4
Resistors 100 Ω	8
Resistors 220 Ω	3
Resistors 1 kΩ	2
Push buttons (2 pins)	8
Male-to-male jumper wires	10 or more

For the power supply part: [DC regulated supply].

3. Background

Diode. A silicon diode conducts current from anode to cathode only when the anode is higher than the cathode by about 0.7 V (forward bias). Otherwise it blocks current (reverse bias). This one-way behavior is what lets us build logic gates from diodes.

Pull-up and pull-down resistors. A pull-down resistor connects a node to ground so it reads a clear LOW when nothing drives it. A pull-up resistor connects it to +5 V so it reads HIGH by default. Without them, the node would be floating and its value would be unpredictable.

LED resistor. The series 100 Ω resistor limits the LED current so the LED is not damaged.

Logic levels. In all circuits, +5 V represents logic 1 and ground represents logic 0. Pressing a button applies +5 V to the input.

4. Part 1: Diode-Based Logic Gates

4.1 OR Gate

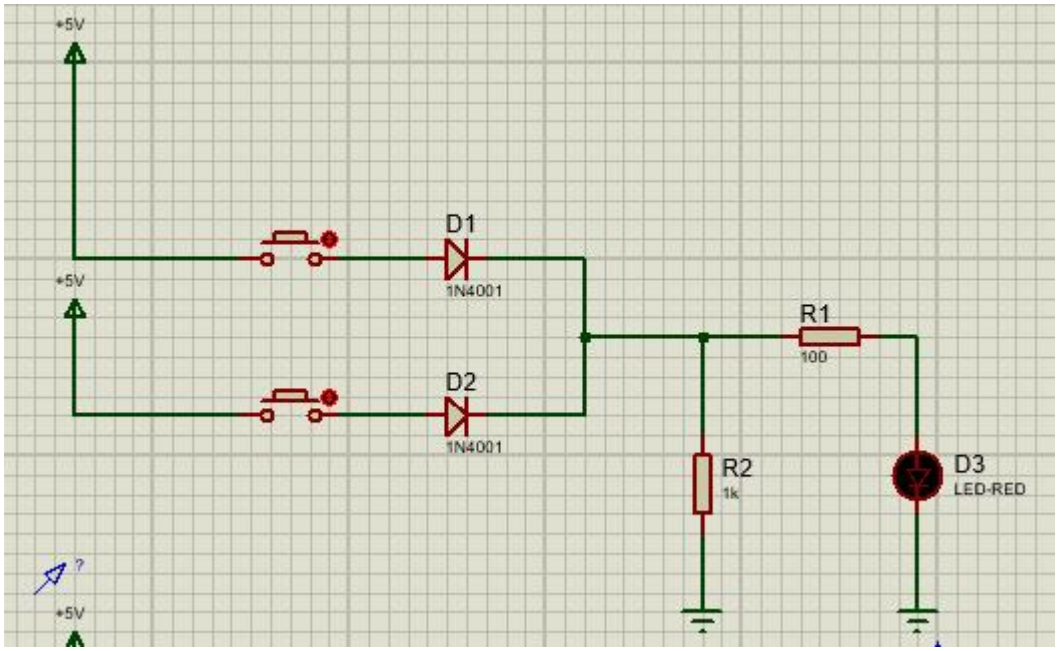
The output is HIGH if at least one input is HIGH.

Circuit. Each button feeds a diode (anode toward the button). The two cathodes are joined at the output node. From this node, a 1 kΩ resistor goes to ground (pull-down) and a 100 Ω resistor in series with the LED goes to ground.

Operation. When a button is pressed, its diode is forward biased and passes +5 V to the output, so the LED turns on. The other diode is reverse biased, so the two inputs do not affect each other. With no button pressed, the 1 kΩ resistor pulls the output to ground and the LED is off. The output is about 5 − 0.7 = 4.3 V when HIGH because of the diode drop.

Truth table:

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	1



Measurements:

A	B	LED
0	0	OFF
0	1	ON
1	0	ON
1	1	ON

4.2 AND Gate

The output is HIGH only when both inputs are HIGH.

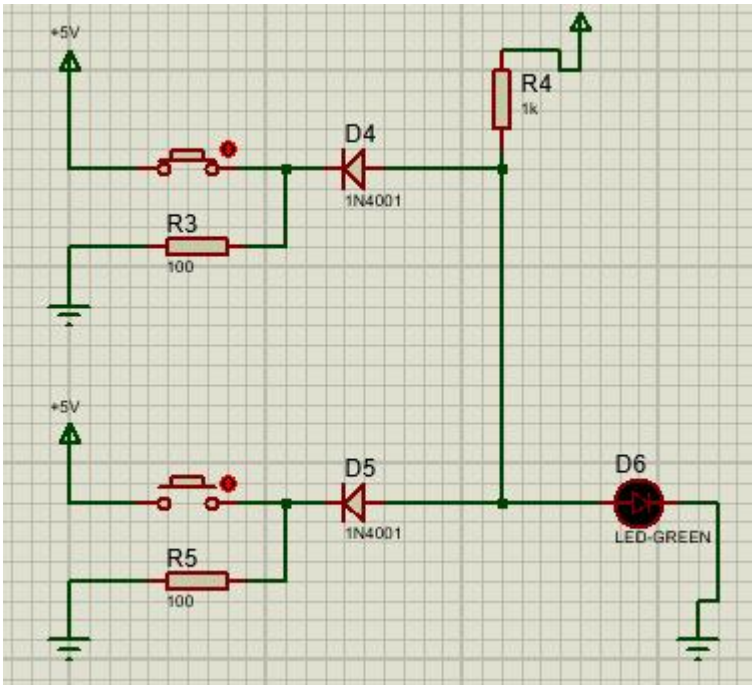
Circuit. A pull-up resistor connects the output node to +5 V. Each input diode is reversed compared with the OR gate (cathode toward the button, anode toward the output). When a button is not pressed, its input is connected to ground through a 100 Ω resistor. The LED is connected from the output node to ground.

Operation. If any input is LOW, its diode is forward biased and pulls the output down to a low voltage, so the LED is off. Only when both buttons are pressed are both diodes reverse biased, nothing pulls the output down, and the pull-up resistor raises it to HIGH, so the LED turns on.

Truth table:

A	B	Output
0	0	0

0	1	0
1	0	0
1	1	1



Measurements:

A	B	LED
0	0	OFF
0	1	OFF
1	0	OFF
1	1	ON

4.3 NOR Gate

The output is HIGH only when both inputs are LOW.

Circuit

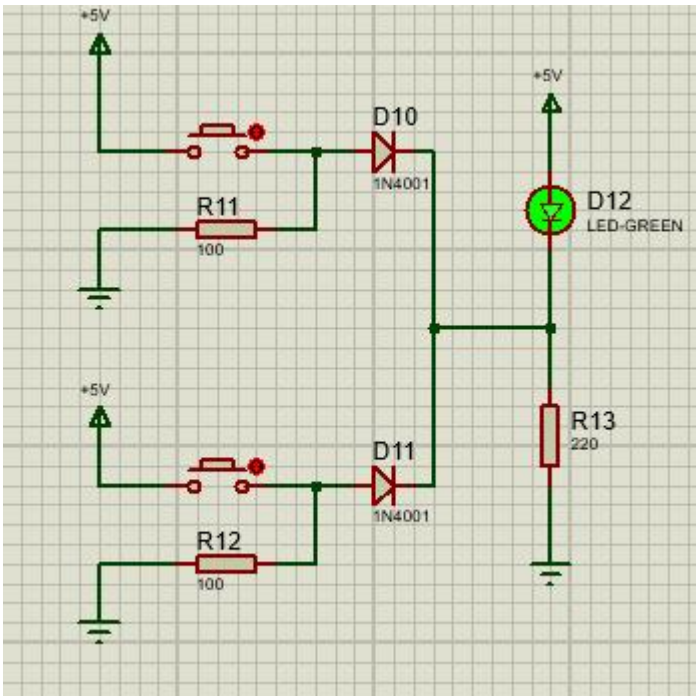
Each input has a push button connected to **+5 V** through a diode (D10 and D11). The cathodes of the two diodes are connected together at the output node. A **220 Ω resistor (R13)** connects the output node to ground, while the LED (D12) is connected between **+5 V** and the output node.

Operation:

When both buttons are released, the output remains **LOW**, allowing current to flow through the LED and R13, so the LED turns **ON**. When either button is pressed, the output rises to about **4.3 V**, reducing the voltage across the LED below its turn-on voltage, so the LED turns **OFF**. Therefore, the circuit produces **NOR logic**.

Truth table:

A	B	Output
0	0	1
0	1	0
1	0	0
1	1	0



Measurements:

A	B	LED
0	0	ON
0	1	OFF
1	0	OFF
1	1	OFF

4.4 NAND Gate

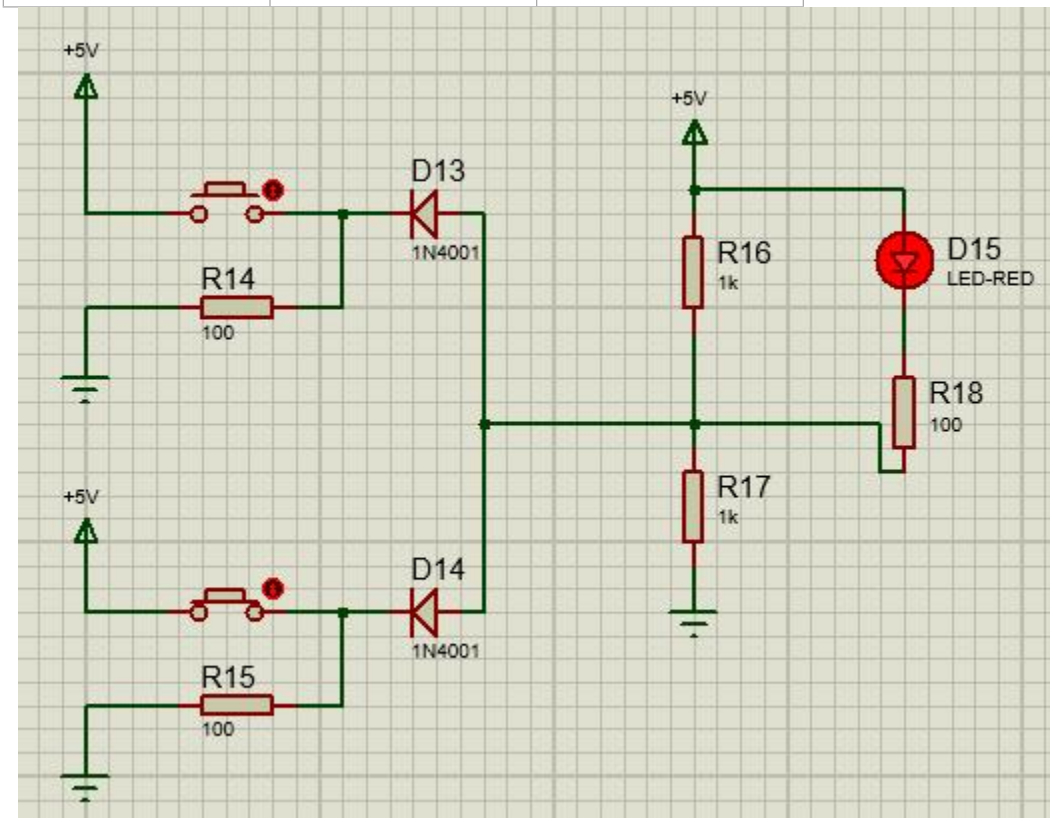
The output is HIGH unless both inputs are HIGH.

Circuit. The input stage is the same as the AND gate: diodes with cathodes toward the buttons and 100 Ω resistors to ground. The output node X is pulled up to +5 V through resistor R8 and pulled down by R9 (1 kΩ). The LED and its 100 Ω series resistor are connected between +5 V and node X.

Operation. Because the LED sits between +5 V and node X, it turns on when X is LOW. If any button is not pressed, a diode pulls X low and the LED is on. When both buttons are pressed, the diodes turn off, X rises, the voltage across the LED drops, and the LED turns off. So the LED shows the inverse of the AND function, which is NAND.

Truth table:

A	B	Output
0	0	1
0	1	1
1	0	1
1	1	0



Measurements:

A	B	LED

0	0	ON
0	1	ON
1	0	ON
1	1	OFF

5. Problems Faced and Solutions

- **Reversed diode.** At first one input diode was placed backwards on the breadboard, so pressing the buttons did nothing. Checking the cathode band (the silver line) and flipping the diode fixed it.
- During the implementation of the **AND gate circuit**, the LED did not operate correctly when using the specified **220 Ω resistor (R6)**. The circuit was tested with the two input buttons, but the LED did not turn on as expected.
  To troubleshoot the issue, the **220 Ω resistor was temporarily replaced with a 1 kΩ resistor**. After making this change, the LED started working correctly

6. Conclusion

We built OR, AND, NOR and NAND gate circuits using diodes and resistors, verified them in Proteus, and implemented them on a breadboard. The truth tables matched the expected behavior. We learned how diode direction decides whether an input can pull the output up or down, why pull-up and pull-down resistors are needed, and why diode logic has limits such as the 0.7 V drop.